

Theorem 5.2.2 Sum of a Geometric Sequence

For any real number r except 1, and any integer $n \geq 0$,

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}$$

Proof (by mathematical induction): Suppose r is a particular but arbitrarily chosen real number that is not equal to 1, and let the property $P(n)$ be the equation

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}$$

We must show that $P(n)$ is true for every integer $n \geq 0$. We do this by mathematical induction on n .

Show that $P(0)$ is true:

To establish $P(0)$, we must show that

$$\sum_{i=0}^0 r^i = \frac{r^{0+1} - 1}{r - 1}$$

The left hand side of this equation is r^0 and the right-hand side is

$$\frac{r^{0+1} - 1}{r - 1} = \frac{r - 1}{r - 1} = 1$$

also because $r^1 = r$ and, since $r \neq 1$, $r - 1 \neq 0$. Hence $P(0)$ is true.

Show that for every integer $k \geq 1$, if $P(k)$ is true then $P(k + 1)$ is also true:

[Suppose that $P(k)$ is true for a particular but arbitrarily chosen integer $k \geq 1$. That is:] Let k be any integer with $k \geq 0$, and suppose that

$$\sum_{i=0}^k r^i = \frac{r^{k+1} - 1}{r - 1}$$

[We must show that $P(k + 1)$ is true. That is:] We must show that

$$\sum_{i=0}^{k+1} r^i = \frac{r^{(k+1)+1} - 1}{r - 1}$$

or, equivalently, that

$$\sum_{i=0}^{k+1} r^i = \frac{r^{k+2} - 1}{r - 1}$$

[We will show that the left-hand side of $P(k + 1)$ equals the right-hand side.]

the left-hand side of $P(k + 1)$ is

$$\begin{aligned} \sum_{i=0}^{k+1} r^i &= \sum_{i=0}^k r^i + r^{k+1} \\ \sum_{i=0}^{k+1} r^i &= \sum_{i=0}^k r^i + r^{k+1} \\ &= \frac{r^{k+1} - 1}{r - 1} + r^{k+1} \\ &= \frac{r^{k+1} - 1}{r - 1} + \frac{r^{k+1}(r - 1)}{r - 1} \\ &= \frac{(r^{k+1} - 1) + r^{k+1}(r - 1)}{r - 1} \\ &= \frac{r^{k+1} - 1 + r^{k+2} - r^{k+1}}{r - 1} \\ &= \frac{r^{k+2} - 1}{r - 1} \end{aligned}$$

Which is the right-hand side of $P(k + 1)$ [as was to be shown.]

[Since we have proved the basis step and the inductive step, we conclude that the theorem is true.]